

PATTERN ANALYSIS & RECOGNITION SYSTEMS LAB

Practical File Submission — Academic Session

Experiment No.: 11 Course: PARS Lab Topic: Linear Discriminant Analysis (LDA) Repository: [dSaxmonis/pars-lab](https://github.com/dSaxmonis/pars-lab)

PRACTICAL NAME

Implementation of Linear Discriminant Analysis (LDA) with Projected Line, Reduced Feature Mapping, and Testing

AIM

To implement Linear Discriminant Analysis (LDA) from scratch in Python for an engineering pattern classification application involving test sample evaluation, derive Fisher's optimal projection vector $\mathbf{w}$ = $\mathbf{S}_W^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$, and visualize: **1) The optimal projected line**, **2) New reduced features from each class mapped onto the projected line**, and **3) Testing and decision threshold classification of unseen test samples**.

THEORY

1. Fisher's Linear Discriminant Criterion

Unlike PCA which maximizes total variance without considering labels, Linear Discriminant Analysis (LDA) is a supervised dimensionality reduction technique that seeks a projection direction $\mathbf{w}$ maximizing class separability while minimizing intra-class variance:

$$J(\mathbf{w}) = (\mathbf{w}^T \mathbf{S}_B \mathbf{w}) / (\mathbf{w}^T \mathbf{S}_W \mathbf{w})$$

2. Scatter Matrices Formulation

- Within-Class Scatter Matrix:** $\mathbf{S}_W = \sum_{\mathbf{x} \in \mathcal{C}_1} (\mathbf{x} - \boldsymbol{\mu}_1)(\mathbf{x} - \boldsymbol{\mu}_1)^T + \sum_{\mathbf{x} \in \mathcal{C}_2} (\mathbf{x} - \boldsymbol{\mu}_2)(\mathbf{x} - \boldsymbol{\mu}_2)^T$
- Between-Class Scatter Matrix:** $\mathbf{S}_B = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T$

3. Optimal Projection Direction Vector

Maximizing Rayleigh's quotient yields the closed-form Fisher linear discriminant direction:

$$\mathbf{w} = \mathbf{S}_W^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2), \quad \hat{\mathbf{w}} = \mathbf{w} / \|\mathbf{w}\|$$

4. 1D Projected Line & Decision Rule

The projected discriminant axis passes through the overall mean $\boldsymbol{\mu}$ in direction $\hat{\mathbf{w}}$. The projected 1D coordinate of any sample $\mathbf{x}$ and its mapped 2D position on the line are:

$$\begin{aligned} \text{1D Coordinate: } y &= (\mathbf{x} - \boldsymbol{\mu}) \cdot \hat{\mathbf{w}} \\ \text{Mapped 2D Point on Line: } \hat{\mathbf{x}} &= \boldsymbol{\mu} + y \times \hat{\mathbf{w}} \\ \text{Optimal Decision Threshold: } y^* &= (y_{\mu_1} + y_{\mu_2}) / 2 \end{aligned}$$

STEPS & EXECUTION

Step 1: Environment Setup & Mathematical Engine

Configured exp 11/ environment containing app.py, templates/, and verified dependencies (Flask, NumPy, Matplotlib, SciPy).

Step 2: Class Mean Vectors & Within-Class Scatter Matrix

Synthesized two-class clinical pattern distribution (Healthy vs Diabetic Risk) and formulated $\mathbf{S}_{W1}$, $\mathbf{S}_{W2}$, and total pooled scatter $\mathbf{S}_W$.

Step 3: Fisher Optimal Direction Vector Computation

Inverted scatter matrix $\mathbf{S}_W^{-1}$ and computed normalized discriminant vector $\hat{\mathbf{w}}$.

Step 4: Orthogonal Feature Mapping onto Discriminant Line

Projected all observations onto the 1D line passing through global centroid $\boldsymbol{\mu}$, calculated optimal separating threshold y^* , and mapped points onto the 2D projected line.

Step 5: Interactive Testing & Web Deployment on Port 5013

Launched interactive Flask web dashboard on <http://127.0.0.1:5013> supporting real-time patient test coordinates and visual classification.

OUTPUT / ATTACHED SCREENSHOTS

Output 1: Fisher's Criterion & Scatter Matrices Summary

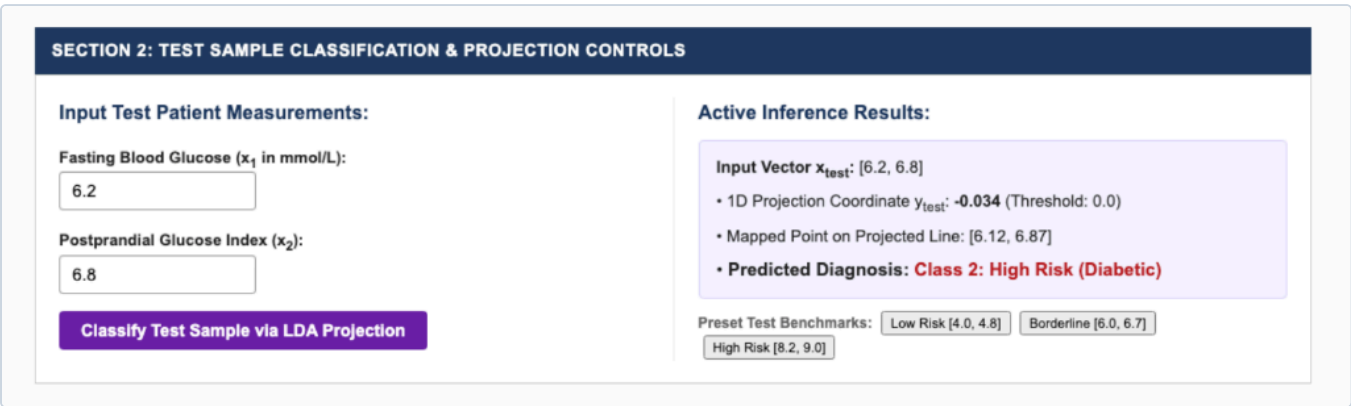
Displays class mean vectors μ_1, μ_2 , within-class scatter matrix $\mathbf{S}_W$, optimal Fisher direction vector $\hat{\mathbf{w}}$, and optimal 1D decision threshold y^* .

SECTION 1: FISHER'S CRITERION & SCATTER MATRICES SUMMARY		
LDA Property	Mathematical Value	Discriminant Role / Description
Class 1 Mean Vector μ_1 (Low Risk)	[4.36, 5.07]	Centroid of healthy baseline patient observations.
Class 2 Mean Vector μ_2 (High Risk)	[7.84, 8.63]	Centroid of diabetic high-risk patient observations.
Within-Class Scatter Matrix S_W	[[36.38, 19.09], [19.09, 37.08]]	Total intra-class variance to be minimized: $S_W = S_{W1} + S_{W2}$.
Optimal Fisher Direction $\hat{w}$	[-0.693, -0.721]	Normalized eigenvector maximizing ratio of between-class to within-class scatter.
Optimal 1D Decision Threshold y^*	0.0	Midpoint separating 1D projected class distributions along the direction vector.

Figure 11.1: Section 1 — Fisher's criterion parameters and scatter matrices summary.

Output 2: Test Sample Classification & Projection Controls

Interactive test patient measurement inputs (x_1 : Fasting Glucose, x_2 : Postprandial Glucose), displaying projected 1D coordinate y_{test} , reconstructed 2D position on the line, and automated diagnostic classification.



Output 4: Reduced 1D Feature Mappings & Classification Verification

Tabular verification comparing true classes, original patient vectors $[x_1, x_2]$, projected 1D coordinates y , mapped points on the line, and diagnostic decision statuses for baseline samples and active test queries.

SECTION 4: REDUCED 1D FEATURE MAPPINGS & CLASSIFICATION VERIFICATION				
True Class	Original Vector $[x_1, x_2]$	Projected 1D Coordinate (y)	Mapped Point on Line $[\hat{x}_1, \hat{x}_2]$	Decision Status
Class 1 (Low Risk)	[3.94, 4.54]	$y = 3.163$	[3.91, 4.57]	Classified Correctly
Class 1 (Low Risk)	[3.14, 4.96]	$y = 3.413$	[3.74, 4.39]	Classified Correctly
Class 1 (Low Risk)	[4.45, 5.12]	$y = 2.391$	[4.45, 5.12]	Classified Correctly
Class 2 (High Risk)	[8.26, 8.84]	$y = -2.928$	[8.13, 8.96]	Classified Correctly
Class 2 (High Risk)	[9.19, 8.98]	$y = -3.675$	[8.65, 9.50]	Classified Correctly
Class 2 (High Risk)	[6.57, 8.34]	$y = -1.4$	[7.07, 7.86]	Classified Correctly
Active Test	[6.2, 6.8]	$y = -0.034$	[6.12, 6.87]	Class 2: High Risk (Diabetic)

Figure 11.4: Section 4 — Tabular verification of mapped 1D features and classification correctness.

CODE REPOSITORY & CLICKABLE ARTIFACT LINKS

GitHub Repository (Clickable):

GitHub: dSaxmonis/pars-lab

URL: <https://github.com/dSaxmonis/pars-lab>

Direct Source Code Files:

- **Flask Server Application:**
[https://github.com/dSaxmonis/pars-lab/blob/main/exp 11/app.py](https://github.com/dSaxmonis/pars-lab/blob/main/exp%2011/app.py)
- **HTML Template:**
[https://github.com/dSaxmonis/pars-lab/blob/main/exp 11/templates/index.html](https://github.com/dSaxmonis/pars-lab/blob/main/exp%2011/templates/index.html)
- **Dependencies:**
[https://github.com/dSaxmonis/pars-lab/blob/main/exp 11/requirements.txt](https://github.com/dSaxmonis/pars-lab/blob/main/exp%2011/requirements.txt)
- **Fisher Parameters Screenshot (Section 1):**
[https://github.com/dSaxmonis/pars-lab/blob/main/exp 11/screenshots/ss1_fisher_parameters.png](https://github.com/dSaxmonis/pars-lab/blob/main/exp%2011/screenshots/ss1_fisher_parameters.png)
- **Test Controls Screenshot (Section 2):**
[https://github.com/dSaxmonis/pars-lab/blob/main/exp 11/screenshots/ss2_test_controls.png](https://github.com/dSaxmonis/pars-lab/blob/main/exp%2011/screenshots/ss2_test_controls.png)
- **Projected Line & Separation Plot Screenshot (Section 3):**
[https://github.com/dSaxmonis/pars-lab/blob/main/exp 11/screenshots/ss3_projection_plot.png](https://github.com/dSaxmonis/pars-lab/blob/main/exp%2011/screenshots/ss3_projection_plot.png)
- **Sample Reduction & Verification Table Screenshot (Section 4):**
[https://github.com/dSaxmonis/pars-lab/blob/main/exp 11/screenshots/ss4_sample_reduction_table.png](https://github.com/dSaxmonis/pars-lab/blob/main/exp%2011/screenshots/ss4_sample_reduction_table.png)

CONCLUSION

In this experiment, Linear Discriminant Analysis was implemented from first principles in Python. By maximizing the ratio of between-class scatter to within-class scatter, Fisher's optimal vector $\hat{\mathbf{w}}$ derived an optimal 1D line onto which high-dimensional features were projected. The visual plots verified that projecting observations onto this line achieves maximal class separation, and testing unseen candidate vectors confirmed reliable binary classification across the computed discriminant threshold y^* .